

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Mathematical modelling provides an important means of representing i,52real-world phenomena using mathematical concepts and structures. Through mathematical models, relationships between different variables can be expressed, analysed and used to make predictions.

One important mathematical modelling technique is **linear regression**. Linear regression is widely used when the objective is to examine the relationship between a dependent variable and one or more independent variables. When more than one independent variable is involved, the method is referred to as **multiple linear regression**.

Multiple linear regression can be expressed naturally using vectors and matrices. Instead of writing a separate equation for every observation, the entire regression model can be represented compactly in matrix form as

$$y = x\beta + \varepsilon,$$

Where

y

Represents the response vector,

x

Represents the design matrix,

β

Represent the vector of the unknown regression coefficient and

ε

Represents the vector of random errors.

The explanatory variables considered include:

- age;
- study hours per day;
- social media hours;
- Netflix hours;
- attendance percentage;
- sleep hours;
- exercise frequency;
- mental health rating;
- gender;
- part-time job status;
- diet quality;
- parental education level;
- internet quality; and

extracurricular participation.

The use of matrices in regression is particularly important from the perspective of mathematics because the estimation of the regression coefficients involves fundamental concepts from linear algebra, including matrix multiplication, transpose, inverse matrices, systems of linear equations, eigenvalues, singular values and matrix conditioning.

Academic performance is affected by several factors. These may include the amount of time a student spends studying,

attendance, sleeping habits, physical exercise, mental health, social media usage, entertainment activities and other demographic or environmental factors.

Understanding the relationship between these variables and examination performance can be useful for identifying factors associated with academic outcomes. However, simply obtaining predictions is not sufficient from a mathematical perspective. It is also important to investigate the numerical behaviour of the matrices used in obtaining those predictions.

For example, the normal-equation approach to least-squares regression requires the matrix $X^T X$.

If this matrix is poorly conditioned, small changes in the data can produce relatively large changes in the estimated coefficients. Consequently, concepts such as the condition number, eigenvalues and perturbation analysis become important. This study therefore combines multiple linear regression with matrix algebra and numerical analysis to predict students' examination scores while examining the mathematical properties and numerical stability of the resulting model.

1.2 Statement of the Problem

Multiple linear regression provides an effective method for predicting a dependent variable from several explanatory variables. However, the computation of regression coefficients

may be affected by the numerical characteristics of the predictor matrix.

When variables are measured on substantially different scales or when there are strong relationships between predictors, the matrix $X^T X$ may become poorly conditioned. A poorly conditioned matrix can make numerical calculations sensitive to small perturbations.

Although many predictive studies focus primarily on prediction accuracy, less attention is often given to the underlying matrix properties that affect the reliability of the computation.

Therefore, there is a need to investigate students' examination-score prediction from both predictive and mathematical perspectives. This study addresses this problem by applying matrix algebra to multiple linear regression and examining prediction accuracy, coefficient sensitivity, perturbation behaviour, eigenvalues and condition numbers.

1.3 Aim of the Study

The main aim of this study is to apply matrix algebra to the development and numerical analysis of a multiple linear regression model for predicting students' examination scores.

1.4 Objectives of the Study

The specific objectives are to:

1. formulate the multiple linear regression model in matrix form;
2. construct an appropriate design matrix from numerical and categorical variables;
3. estimate regression coefficients using the least-squares method;
4. evaluate the predictive performance of the regression model;
5. determine the relative influence of the explanatory variables;
6. perform sensitivity and perturbation analysis;
7. investigate the eigenvalues and condition numbers of the predictor matrices;
8. examine the effect of standardization on numerical conditioning;
9. compare different methods for solving the least-squares system; and
10. demonstrate the practical application of matrix algebra in academic-performance prediction.

1.5 Research Questions

The study seeks to answer the following questions:

1. How can multiple linear regression be represented and solved using matrix algebra?
2. How accurately can the selected variables predict students' examination scores?
3. Which variables have the greatest influence on the predicted examination scores?

4. How does standardization affect the numerical conditioning of the design matrix?
5. How sensitive are the regression coefficients and predictions to small perturbations?
6. Do the inverse, direct-solve and pseudoinverse methods produce consistent regression estimates?

1.6 Significance of the Study

This study is significant because it demonstrates a practical application of mathematical concepts that are central to linear algebra and numerical analysis. The study demonstrates how matrices can be used to:

- represent a multiple regression model;
- estimate unknown parameters;
- analyse relationships between variables;
- investigate numerical stability;
- examine sensitivity to perturbations;
- analyse eigenvalues and condition numbers; and
- obtain predictions from a mathematical model.

The study may also be useful to students and researchers interested in the application of linear algebra, numerical methods, mathematical modelling and computational mathematics.

1.7 Scope of the Study

The study focuses on predicting students' examination scores using multiple linear regression.

The study specifically focuses on matrix-based least-squares estimation, model evaluation, sensitivity analysis, perturbation analysis, eigenvalues, condition numbers and standardization.

1.8 Definition of Key Terms

Matrix

A matrix is a rectangular arrangement of numbers or mathematical elements organized into rows and columns.

Design Matrix

The design matrix is the matrix containing the observations of the explanatory variables, including an intercept column where applicable.

Multiple Linear Regression

Multiple linear regression is a statistical and mathematical modelling technique used to predict a dependent variable using two or more explanatory variables.

Least Squares

Least squares is an estimation method that determines the coefficients that minimize the sum of squared differences between observed and predicted values.

Residual

A residual is the difference between an observed value and its corresponding predicted value.

Eigenvalue

An eigenvalue is a scalar associated with a square matrix that describes the scaling factor of an eigenvector under matrix transformation.

Condition Number

The condition number measures the sensitivity of a matrix-related problem to small changes or perturbations in the input.

Standardization

Standardization is the transformation of variables to a common scale, usually by subtracting the mean and dividing by the standard deviation.